

Chapter 6: The Normal Distribution

Sections 6.1–6.4

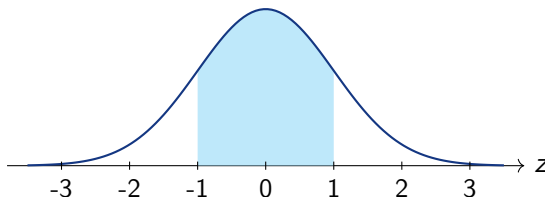
MAT 1272 Statistics

Today's Main Idea

The Big Picture

For a normal distribution, probability is represented by area under a bell-shaped curve.

Probability = Area under the curve



6.1 Continuous Random Variables

Discrete vs. Continuous

Discrete: values can be counted.

Examples: number of students, number of heads, number of cars.

Continuous: values are measured on an interval.

Examples: height, time, weight, temperature.

Important

For a continuous random variable,

$$P(X = a) = 0.$$

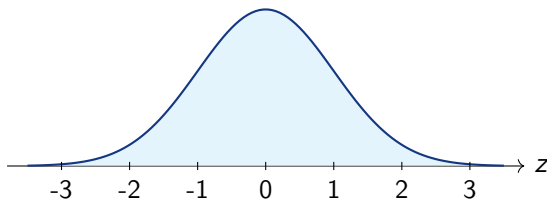
We find probabilities using intervals, such as $P(a < X < b)$.

6.1 Normal Distribution

Main Characteristics

A normal distribution is:

- Bell-shaped
- Symmetric
- Centered at the mean
- Has total area equal to 1



Mean and Standard Deviation

Parameters

A normal distribution is described by two numbers:

$$\mu = \text{mean}$$

$$\sigma = \text{standard deviation}$$

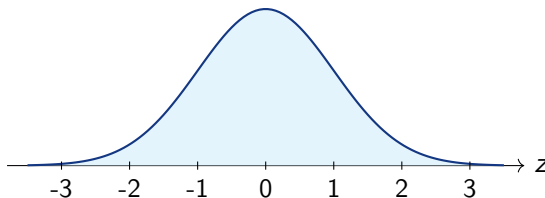
- The mean μ controls the center.
- The standard deviation σ controls the spread.
- Larger σ means a wider curve.

The Standard Normal Distribution

Definition

The standard normal distribution has

$$\mu = 0 \quad \text{and} \quad \sigma = 1.$$



$$Z \sim N(0, 1)$$

The Empirical Rule

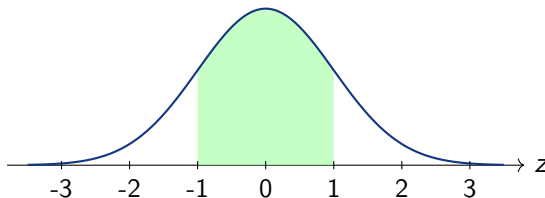
68–95–99.7 Rule

For a normal distribution:

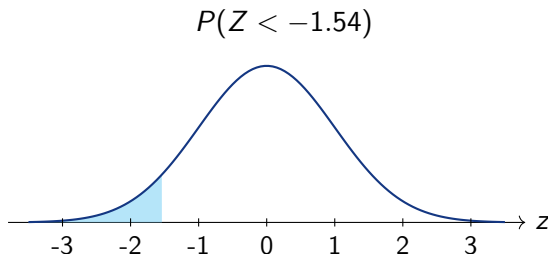
68% is within 1 standard deviation

95% is within 2 standard deviations

99.7% is within 3 standard deviations



Area to the Left

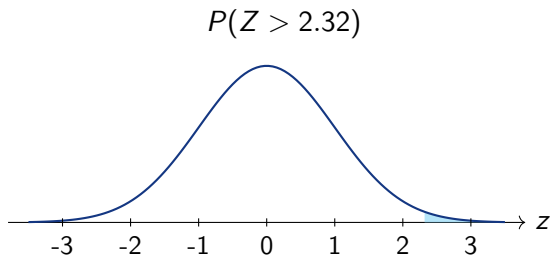


Calculator

```
normalcdf(-1E99,-1.54,0,1)
```

$$P(Z < -1.54) \approx 0.0618$$

Area to the Right



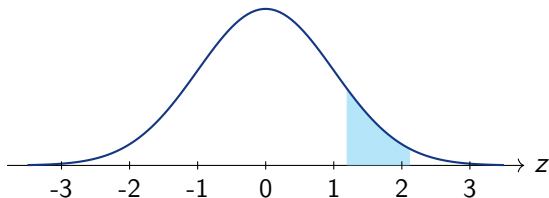
Calculator

```
normalcdf(2.32,1E99,0,1)
```

$$P(Z > 2.32) \approx 0.0102$$

Area Between Two Values

$$P(1.19 < Z < 2.12)$$



Calculator

```
normalcdf(1.19,2.12,0,1)
```

$$P(1.19 < Z < 2.12) \approx 0.1000$$

6.2 Standardizing a Normal Distribution

The Key Formula

To convert an x -value into a standard normal value:

$$z = \frac{x - \mu}{\sigma}$$

- x = observed value
- μ = mean
- σ = standard deviation
- z = number of standard deviations from the mean

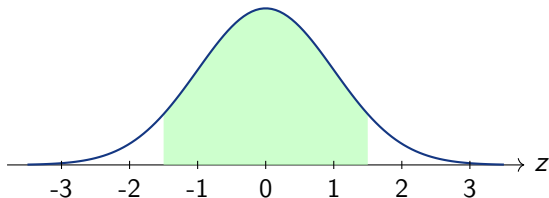
Interpretation

Positive z : above the mean

Negative z : below the mean

Understanding z-Scores

$$z = \frac{x - \mu}{\sigma}$$



- $z = 0$: exactly at the mean
- $z = 1$: one standard deviation above the mean
- $z = -2$: two standard deviations below the mean

Example 6-6

A normal distribution has

$$\mu = 50, \quad \sigma = 10$$

Find the z-score for $x = 55$.

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A normal distribution has

$$\mu = 50, \quad \sigma = 10$$

Find the z-score for $x = 55$.

$$z = \frac{55 - 50}{10}$$

$$z = \frac{5}{10}$$

$$z = 0.50$$

Example 6-6

A normal distribution has

$$\mu = 50, \quad \sigma = 10$$

Find the z-score for $x = 55$.

$$z = \frac{55 - 50}{10}$$

$$z = \frac{5}{10}$$

$$z = 0.50$$

$$P(X < 55) = P(Z < 0.50)$$

$$= .6915$$

Example 6-6 (Part b)

$$\mu = 50, \quad \sigma = 10$$

Find the z-score for $x = 35$.

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$$\mu = 50, \quad \sigma = 10$$

Find the z-score for $x = 35$.

$$z = \frac{35 - 50}{10}$$

$$z = -1.50$$

Example 6-6 (Part b)

$$\mu = 50, \quad \sigma = 10$$

Find the z-score for $x = 35$.

$$z = \frac{35 - 50}{10}$$

$$z = -1.50$$

$$P(X < 35) = P(Z < -1.50)$$

$$= .0668$$

Example 6-7

Suppose

$$\mu = 25, \quad \sigma = 4$$

Find

$$P(25 < X < 32)$$

Example 6-7

Suppose

$$\mu = 25, \quad \sigma = 4$$

Find

$$P(25 < X < 32)$$

Convert both values to z-scores.

$$z_1 = \frac{25 - 25}{4} = 0$$

$$z_2 = \frac{32 - 25}{4} = 1.75$$

Example 6-7

Suppose

$$\mu = 25, \quad \sigma = 4$$

Find

$$P(25 < X < 32)$$

Convert both values to z-scores.

$$z_1 = \frac{25 - 25}{4} = 0$$

$$z_2 = \frac{32 - 25}{4} = 1.75$$

$$P(25 < X < 32) = P(0 < Z < 1.75)$$

$$= .9599 - .5000$$

Example 6-7 (Different Sides of Mean)

Find

$$P(18 < X < 34)$$

with

$$\mu = 25, \quad \sigma = 4$$

Example 6-7 (Different Sides of Mean)

Find

$$P(18 < X < 34)$$

with

$$\mu = 25, \quad \sigma = 4$$

$$z_1 = \frac{18 - 25}{4} = -1.75$$

$$z_2 = \frac{34 - 25}{4} = 2.25$$

Example 6-7 (Different Sides of Mean)

Find

$$P(18 < X < 34)$$

with

$$\mu = 25, \quad \sigma = 4$$

$$z_1 = \frac{18 - 25}{4} = -1.75$$

$$z_2 = \frac{34 - 25}{4} = 2.25$$

$$P(-1.75 < Z < 2.25)$$

$$= .9878 - .0401$$

TI-84: Normal Probabilities

Finding areas under a normal curve

Press:

2nd

VARs

Select

2:normalcdf(

Calculator format:

`normalcdf(lower,upper,mu,sigma)`

TI-84 Example

Find

$$P(18 < X < 34)$$

for

$$\mu = 25, \quad \sigma = 4$$

Enter:

```
normalcdf(18,34,25,4)
```

Calculator gives:

0.9477

6.3 Applications of the Normal Distribution

Strategy

Word problems always follow the same steps:

- 1 Draw a picture.
- 2 Identify μ and σ .
- 3 Convert x to z .
- 4 Find probability.

Example 6-11: Physician Salaries

$$\mu = 196,000$$

$$\sigma = 20,000$$

Find

$$P(169,400 < X < 206,800)$$

Example 6-11: Physician Salaries

$$\mu = 196,000$$

$$\sigma = 20,000$$

Find

$$P(169,400 < X < 206,800)$$

$$z_1 = \frac{169400 - 196000}{20000} = -1.33$$

$$z_2 = \frac{206800 - 196000}{20000} = .54$$

Example 6-11 Continued

$$P(-1.33 < Z < .54)$$

Example 6-11 Continued

$$P(-1.33 < Z < .54)$$

$$= .7054 - .0918$$

$$= .6136$$

Example 6-11 Continued

$$P(-1.33 < Z < .54)$$

$$= .7054 - .0918$$

$$= .6136$$

$$61.36\%$$

of physicians earn between those values.

Example 6-12: Toy Assembly

$$\mu = 55$$

$$\sigma = 4$$

Find

$$P(X \leq 60)$$

Example 6-12: Toy Assembly

$$\mu = 55$$

$$\sigma = 4$$

Find

$$P(X \leq 60)$$

$$z = \frac{60 - 55}{4} = 1.25$$

Example 6-12: Toy Assembly

$$\mu = 55$$

$$\sigma = 4$$

Find

$$P(X \leq 60)$$

$$z = \frac{60 - 55}{4} = 1.25$$

$$P(X \leq 60) = P(Z \leq 1.25)$$

$$= .8944$$

Example 6-13: Soda in a Can

$$\mu = 12$$

$$\sigma = .015$$

Find

$$P(11.97 < X < 11.99)$$

Example 6-13: Soda in a Can

$$\mu = 12$$

$$\sigma = .015$$

Find

$$P(11.97 < X < 11.99)$$

$$z_1 = \frac{11.97 - 12}{.015} = -2.00$$

$$z_2 = \frac{11.99 - 12}{.015} = -0.67$$

Example 6-13 Continued

$$P(-2.00 < Z < -0.67)$$

Example 6-13 Continued

$$P(-2.00 < Z < -0.67)$$

$$= .2514 - .0228$$

$$= .2286$$

Example 6-13 Continued

$$P(-2.00 < Z < -0.67)$$

$$= .2514 - .0228$$

$$= .2286$$

$$22.86\%$$

Class Practice

A distribution has

$$\mu = 40$$

$$\sigma = 5$$

Find:

- ① $P(X > 55)$
- ② $P(X < 49)$
- ③ $P(35 < X < 45)$

Use your TI-84.

6.4 Working Backwards

New Goal

So far we have done:

$$x \longrightarrow z \longrightarrow \text{Probability}$$

Now we reverse the process:

$$\text{Probability} \longrightarrow z \longrightarrow x$$

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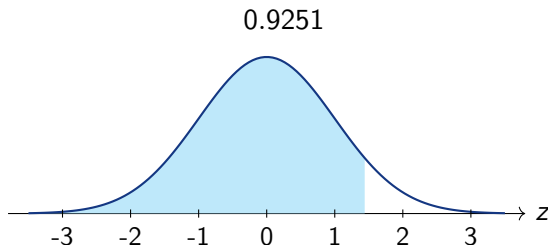
$$\text{Probability} \longrightarrow z \longrightarrow x$$

Examples:

- Find the z-score corresponding to the top 5%.
- Find the exam score that separates the top 10%.
- Find the warranty period for a product.

Finding a z-Value from an Area

Suppose the area to the left is



We seek the z-score whose cumulative area is 0.9251.

$$z \approx 1.44$$

Example 6-15

Find the z-score such that

$$P(Z < z) = 0.9251$$

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Using the z-table:

$$0.9251$$

corresponds to

$$z = 1.44$$

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Using the z-table:

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$$z = 1.44$$

$$P(Z < 1.44) = 0.9251$$

Example 6-16

Find the z-score such that

$$P(Z > z) = 0.0050$$

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$$P(Z > z) = 0.0050$$

Convert to a left-tail area:

$$1 - 0.0050 = 0.9950$$

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Find the z-score such that

$$P(Z > z) = 0.0050$$

Convert to a left-tail area:

$$1 - 0.0050 = 0.9950$$

Find 0.9950 in the z-table.

$$z \approx 2.58$$

Example 6-16

Find the z-score such that

$$P(Z > z) = 0.0050$$

Convert to a left-tail area:

$$1 - 0.0050 = 0.9950$$

Find 0.9950 in the z-table.

$$z \approx 2.58$$

Answer:

$$P(Z > 2.58) \approx 0.005$$

Example 6-17

Find the z-score such that

$$P(Z < z) = 0.05$$

Example 6-17

Find the z-score such that

$$P(Z < z) = 0.05$$

Since the area is less than 0.50, z must be negative.

$$z \approx -1.65$$

Example 6-17

Find the z-score such that

$$P(Z < z) = 0.05$$

Since the area is less than 0.50, z must be negative.

$$z \approx -1.65$$

$$P(Z < -1.65) \approx 0.05$$

Finding z when the area is known

Press

2nd

VARs

Choose

3:invNorm(

Calculator format:

invNorm(area)

for the standard normal distribution.

TI-84 Example

Find the z-score such that

$$P(Z < z) = 0.9251$$

Enter:

```
invNorm(.9251)
```

Calculator returns

$$z \approx 1.44$$

Finding x Values

Once z is known, use

$$x = \mu + z\sigma$$

Finding x Values

Once z is known, use

$$x = \mu + z\sigma$$

This formula converts

z

back into

x

Remember

This is the reverse of

$$z = \frac{x - \mu}{\sigma}$$

Example 6-18: Calculator Warranty

Suppose

$$\mu = 54$$

months

$$\sigma = 8$$

months

Only 1% should fail before warranty expiration.

$$P(X < x) = 0.01$$

Example 6-18 Continued

First find z .

$$P(Z < z) = 0.01$$

$$z \approx -2.33$$

Example 6-18 Continued

First find z .

$$P(Z < z) = 0.01$$

$$z \approx -2.33$$

Now compute x .

$$x = \mu + z\sigma$$

$$x = 54 + (-2.33)(8)$$

$$x = 35.36$$

Example 6-18 Continued

First find z .

$$P(Z < z) = 0.01$$

$$z \approx -2.33$$

Now compute x .

$$x = \mu + z\sigma$$

$$x = 54 + (-2.33)(8)$$

$$x = 35.36$$

Warranty period:

35 months

Example 6-19: Physician Income

Suppose

$$\mu = 196,000$$

$$\sigma = 20,000$$

Only 10% earn more than Dr. Garcia.

Example 6-19: Physician Income

Suppose

$$\mu = 196,000$$

$$\sigma = 20,000$$

Only 10% earn more than Dr. Garcia.

$$P(X > x) = 0.10$$

Example 6-19 Continued

Convert to left-tail area.

$$1 - .10 = .90$$

Example 6-19 Continued

Convert to left-tail area.

$$1 - .10 = .90$$

Find z .

$$z \approx 1.28$$

Example 6-19 Continued

Convert to left-tail area.

$$1 - .10 = .90$$

Find z .

$$z \approx 1.28$$

Now compute x .

$$x = 196000 + 1.28(20000)$$

$$x = 221600$$

Example 6-19 Continued

Convert to left-tail area.

$$1 - .10 = .90$$

Find z .

$$z \approx 1.28$$

Now compute x .

$$x = 196000 + 1.28(20000)$$

$$x = 221600$$

$221,600$

So Dr. Garcia earns about \$221,600.

TI-84 Finding x Directly

Use

```
invNorm(area,mu,sigma)
```

Examples:

```
invNorm(.90,196000,20000)
```

returns

221600

TI-84 Finding x Directly

Use

```
invNorm(area,mu,sigma)
```

Examples:

```
invNorm(.90,196000,20000)
```

returns

221600

No z-table needed!

Exam Strategy

For every normal distribution problem:

- 1 Draw a picture.
- 2 Identify μ and σ .
- 3 Decide whether area is left, right, or between.
- 4 Use:

`normalcdf`

for probabilities.

- 5 Use:

`invNorm`

for percentiles and cutoffs.

Calculator Summary

Probability from x-values

```
normalcdf(lower,upper,mu,sigma)
```

Calculator Summary

Probability from x-values

```
normalcdf(lower,upper,mu,sigma)
```

Find x or z from area

```
invNorm(area,mu,sigma)
```

Calculator Summary

Probability from x-values

```
normalcdf(lower,upper,mu,sigma)
```

Find x or z from area

```
invNorm(area,mu,sigma)
```

Examples:

```
normalcdf(18,34,25,4)
```

```
invNorm(.90,196000,20000)
```

Chapter 6 Review

- 1 Continuous random variables
- 2 Normal distributions
- 3 Standard normal distribution
- 4 Empirical Rule
- 5 z-scores
- 6 Areas under the curve
- 7 Standardizing:

$$z = \frac{x - \mu}{\sigma}$$

- 8 Finding probabilities with
- 9 Finding cutoffs with

`normalcdf`

`invNorm`

Chapter 6 Complete

Questions?